

Creating Generalizable Data-Driven Approaches for Biodiversity Monitoring via Acoustics

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Abstract. Global biodiversity is declining at unprecedented rates, yet traditional monitoring at the necessary scales remains costly and biased toward what can be seen. Sound offers a complementary lens: many species are detected more reliably by their vocalizations, microphones are inexpensive and unobtrusive, and they can cover greater spatial and temporal scales. These advantages have made passive acoustic monitoring a fast-growing paradigm, yet robust, generalizable sound distinction in complex soundscapes remain a central obstacle. My thesis addresses this by combining data-driven human-inspired representation learning with knowledge-guided unsupervised learning from auditory scene analysis and ecological reasoning, prioritizing hierarchical organization and structure discovery prior to labelling. Human-in-the-loop oversight is incorporated as targeted verification under uncertainty, drawing on active learning and weak supervision to direct effort where it has the highest value.

Keywords: Biodiversity monitoring · Clustering validity · Ecoacoustics · Unsupervised learning

1 Introduction

Global biodiversity is undergoing decline at unprecedented rates, necessitating scalable and continuous monitoring solutions beyond traditional, often limited and bias, manual survey methods [1]. Passive Acoustic Monitoring (PAM) has emerged, and now the deployment of inexpensive, non-invasive microphones can continuously capture entire soundscapes [8]. Continental-scale initiatives, such as the Australian Acoustic Observatory (A2O), and the U.S. Northeast Passive Acoustic Sensing Network (NEPAN) exemplify this success, providing wide spatial and temporal coverage crucial for comprehensive ecological assessment [6].

This technological success, however, has created a severe technological bottleneck. The sheer volume of data renders manual analysis impossible. Existing machine learning and deep learning solutions, particularly supervised models, perform sub-optimally in this complex domain because they are typically trained on small, highly sanitized, and species-specific datasets [7]. Thus, these models

are useful for narrow-scope research questions but cannot handle the complications of real-world heterogeneous soundscapes nor generalize beyond the circumscription of the dataset upon which they are trained. Consequently, existing approaches may suffer from severe domain shift when applied across diverse ecological contexts, rendering the scarcity of robust, labelled data a primary obstacle [2].

The core thrust of my doctoral research addresses this impasse by proposing a perspective shift: instead of pursuing premature classification with inadequate labels, I propose prioritizing unsupervised structure discovery prior to labelling. My work investigates scalable frameworks that first reveals the inherent hierarchical organization of soundscapes by treating them as compositions of events, in a manner that is also interpretable and meets the needs of domain experts. Two key issues facing my current research can thus be summarized.

- RQ1: How can I develop a scalable unsupervised framework to automatically discover the compositional structure of complex, multi-source soundscapes, thereby breaking the manual annotation bottleneck?
- RQ2: How can an unsupervised framework be designed to create acoustically pure and ecologically coherent groupings of sound events, particularly in the presence of polyphony, by implementing domain-adaptive criteria for event windowing and clustering?

2 Current Progress

Initial progress involved establishing a lightweight and robust core unsupervised architecture capable of structuring voluminous raw, heterogeneous soundscape data across diverse geographic regions. This addresses the challenge of building a robust scalable framework (RQ1) despite inevitable environmental complexity and domain shift.

I first validated the Mel-Frequency Cepstral Coefficient (MFCC) features for their ability to distinguish between major ecological sound categories (biophony, geophony, anthrophony, silence), achieving exceptional accuracy (98%) across two ecologically distinct sites with a simple classifier [4]. This proved that computationally inexpensive, human-inspired features were sufficient for the task and the complete UMAP-HDBSCAN pipeline successfully organized a large dataset from six biodiverse soundscapes into well-formed clusters.

To resolve the primary annotation bottleneck, I operationalized this framework and thus created LEAVES (Large-Scale Ecoacoustics Analysis and Visualization with Efficient Segmentation), an open-source, human-in-the-loop decision support tool [5]. LEAVES uses the discovered acoustic clusters to enable targeted expert verification, where a label from a small sample is propagated to an entire cluster. This method proved to be up to 7.12 times faster than traditional manual labeling while maintaining around 79-90% similarity to ground truth.

Objective	Timeline
RQ1: Scalable Framework	Feb 2022–Mar 2024
RQ2: Ecologically Coherent Groupings	Mar 2024–Mar 2026
2.1. Adaptive Sound Event Detection	Mar 2024–Feb 2025
2.2. Source Separation	Feb 2025–May 2026
Thesis Writing & Defense	May 2026–August 2026

Table 1. Research Timeline

3 Future Work and Research Timeline

Future work, as shown in Table 1, will address another critical limitation in soundscape analysis: the reliance on fixed-length window sizes. This approach is suboptimal as it causes call fragmentation, and even state-of-the-art deep learning models continue to struggle with variable-length, rapidly shifting vocalizations common in nature [3]. To overcome this, I will develop an adaptive, content-aware, lightweight, and generalisable method for sound event detection. The core strategy involves exploring embedding drift to create an adaptive windowing system that dynamically aligns segmentation with the start and end of actual acoustic events, ensuring their temporal integrity. This refined segmentation is the first step toward solving the more complex problem of separating overlapping sounds in the frequency domain. Current source separation techniques are often inadequate for ecological data as they are computationally expensive, assume a fixed number of sound sources, or fail to generalize across the noisy, nonstationary conditions of unseen environments. Thus, my next steps, if successful, would address this.

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